Lab 03 | Hunting in Memory

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The analysis of this volatile memory (vmem) file mainly employed the use of Volatility. Volatility is a data analyzer that allows researchers to discover data artefacts from vmem samples. I analyzed the file in question, KobayashiMaru.vmem, for malicious activity, and this lab report will summarize my findings and present a conclusion to whether there was indeed any such activity.

To begin, this computer was running the Windows XP SP2x86 operating system and the size of the .vmem file is the amount of RAM that was used in the collection of the information, which was (1a). I was able to determine this from using the command "volatility -f KobayashiMaru.vmem imageinfo". This command printed out a profile that Volatility believes fits the operating system in question (1b). What is also interesting is the command, "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 pslist" printed out the processes that were running at the time of the capture, one of which stood out to me. After doing some research on process names that looked strange, like this one, I was able to discover that the process "hxdef100.exe" is a rootkit and is a very strong sign that something was amiss on this device (2). There was also another suspicious process running called "iroffer.exe", which upon doing research, I found that it was malware that creates backdoors onto systems for easier unauthorized access (2). To continue, I found another likely suspect in this .vmem file: "poisonivy.exe". This is a remote access/Trojan malware that has been extensively documented and has the capability to establish a keylogger, rootkit, and create an encrypted communications channel with the attacker that can be used to exfiltrate data (3).

I continued my investigation by checking the process trees, or what processes are running other processes with the command, "volatility -f KobayashiMaru.vmem profile=WinXPSP2x86 pstree". I started by checking through the most suspicious processes mentioned above and found that some of them did indeed run subprocesses. The process "hxdef100.exe" for example, was running "cryptcat.exe" and "bircd.exe" (4). "cryptcat.exe" is an offshoot "netcat" process that communicates and encrypts data across networks and was likely used to link the attacker with the infected device. "bircd.exe" is short before "beware ircd", which is a process that creates an Internet Relay Chat (IRC) server, which was used in tandem with "cryptcat.exe" to set up the secure communication channel. One process set up the actual transferring of data between the two devices and the other set up the server to be able to exfiltrate data to, grab files to escalate the attacker's access further, and stage more malware onto the infected device. It is also important to note that some of these processes, like "cryptcat.exe", are native to Linux, insinuating the use of a Linux machine to attack and infect this device. Another interesting find using this command was the fact that "explorer.exe", a critical Windows process, was being used to run "poisonivy.exe", the Trojan malware (5). The attacker was quite holistic in his approach to using multiple processes, offensive tools, and backdoors to persistently keep this device in his grasp.

Following this, I checked for any processes that were hiding using the command "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 psxview" (6)(7). The first two columns, "pslist" and "psscan", alert us to the existence of processes that are running and have run on the system. If you see a process that shows as "False" in one and/or both columns, the process is attempting to hide itself, and in our case, all the processes in (6) and (7) are doing so. I was able to conclude this because the processes were shown as "False" in the "psscan" column. Processes that I already determined were suspicious showed up here once again, namely "cryptcat.exe", "bircd.exe", "poisonivy.exe", "iroffer.exe", and "hxdef100.exe". Since I knew that some of these processes were used to facilitate communication between the infected device and the attacker's device, it made sense to me to check what addresses were communicating with the infected device at the time of the capture.

To find these addresses, I employed a few commands. The first one was, "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 connscan". This scans the .vmem file for any connections made to other addresses. I was able to find 3 in total, but these were either loopback addresses or a local address, so it did not particularly help me further in finding what addresses were connecting to the device (8). The silver lining, though, was that the process IDs for these connections directly correlated with the process IDs for some of the suspicious processes. 1728 matched with "iroffer.exe", 1480 matched with "bircd.exe", and 480 matched with "poisonivy.exe". This makes sense since these are processes that are either establishing connection and data exfiltration channels between the infected device and the attacking device or enabling easier connection between them. Next, I used the commands "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 sockets" and "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 netscan". I believed these commands yielded no results at the time, so the hunt just continued from there. I will get back to what I found later. I had good leads with these 3 processes so far, but I figured there were probably more that went under the radar when I first looked at all the processes in the first command. I went off of a pure judgment call and mainly researched the most suspicious looking ones before anything else. My criteria for that were looking up any English words, like in "poisonivy.exe", and looking for unfamiliar shortenings like "bircd.exe" and any other processes I did not immediately recognize as processes that should be running on a Windows device. This is why I did not investigate "svchost.exe" and "rundll32.exe" until the following section. This is where I began to see a broader picture of the scope of the infection.

I ran the command "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 cmdline" to see what commands were run on the device in the command line, and here I found plenty of evidence of tampering and found the hack was deeper than I first believed. Screenshots (9-13) contain evidence of things like hidden directories being used, suspicious DLL's being run, "svchost.exe" being used to check the running services on the device (evidence of enumeration),

communication with the bircd server using previously established suspicious processes, and more. There was almost certainly data exfiltration occurring since FTP was being used in conjunction with netcat (nc), which allows for data transfer between devices (13). I was also able to identify new process ID's and ascertain the full paths to multiple directories, processes, and files that were involved in the infection.

The next big breakthrough in the investigation came about when using the commands "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 procdump -p 480 –dump-dir ." and "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 memdump -p 480 –dump-dir ." This allows us to look closer at what happened in the memory itself when the "poisonivy.exe" process was running (the process associated with the ID 480". This was also the only process that I discovered earlier that was associated with an address that was not just a loopback address. I also used the commands "strings 480.dmp | grep -Fi "192.168.5.98" -C 5" to find any mention of this address in the file, then used the command "strings 480.dmp | less" to take a look at the individual pages of information. I was able to find something interesting, a username. Daniel Faraday (14). This is the username of the victim's device, since in the dump, we can see after "poisonivy.exe" is used that this user folder in the Windows device is being searched through extensively (15). "poisonivy.exe" is the Trojan malware, that much was known, but what was unknown was what payload was used to deliver it to the system in the first place? This is a hiding process that is already difficult enough to remove on its own, but no matter how many times you remove it, if you do not remove the process that installs it, it will come back endlessly. Considering I knew this system was infected at the root level because of the fact that a backdoor and a rootkit were installed in this from previous information, I took my search to the registry keys.

To look through the registry keys, I used the command "volatility -f KobayashiMaru.vmem –profile=WinXPSP2x86 printkey -K (key name)" and began by checking the Run key. This key would allow me to check for what would begin to run on startup and it led me to look back at a process I had only barely considered to be suspicious at the start. This process was "msmsgs.exe", the Windows messenger process, which was on startup under Daniel Faraday's user (15-17). I conclude that this was the original payload that allowed the attacker to gain access to the victim's system and is, at the very least, necessarily linked to "poisonivy.exe", the Trojan malware, since it allowed Poison Ivy to be installed on the system. I was able to confirm this further by using the command "strings 480.dmp | grep -Fi "msmsgs.exe", which searches inside my dumps of the process ID 480 for any mention of "msmsgs.exe", and there were, indeed, 2 uses of it inside (17). Before I give my full summary and conclusion, I would like to take a step back and answer the following unanswered questions.

There were many suspicious file paths discovered throughout my investigation, and I will name them specifically here. “hxdef100.exe” was running in “C:\hxdefrootkit\hxdef100.exe”, “iroffer.exe” was running in “C:\hidden\ir\iroffer.exe”, “bircd.exe” was running in “C:\hidden\bewareircd-win32\bircd.exe”, and “poisonivy.exe” was running in “C:\WINDOWS\System32\poisonivy.exe”. An honorable mention goes to one suspicious batch file ,“lock.bat”, running in “C:\inetpub\Ftproot\lock.bat” (18). What interested me about this is they also used the /K switch when they ran this file, meaning they kept a command line window open after running it to check its output. One of the most telling commands that I have yet to mention was this “netcat” command: “-L -p 6666 -e cmd.exe”. This command sets up the backdoor. “-L” means to listen persistently, meaning new connections will be attempted should the original one end. “-p 6666” means the command will listen on port 6666, a TCP port. “-e cmd.exe” opens a command line to interact with the device.

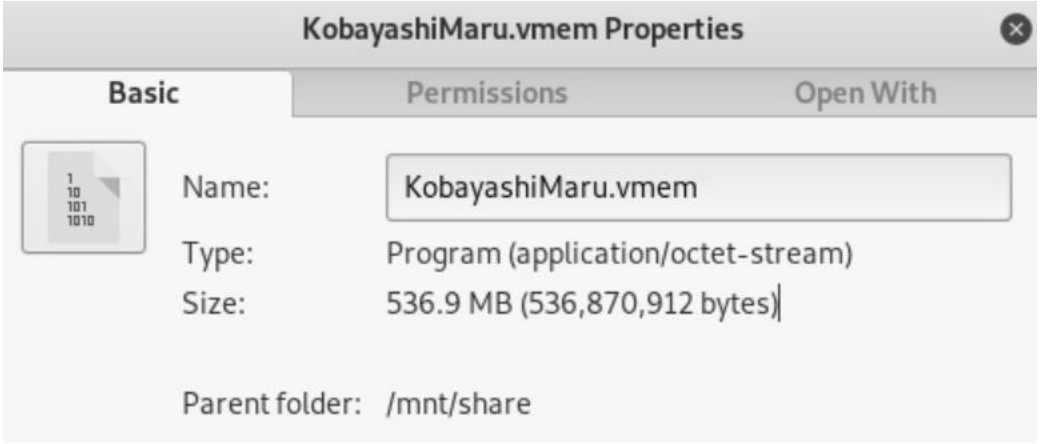
To conclude my investigation, this box was royally hacked. A payload was first introduced, that being “msmsgs.exe”, which allowed “poisonivy.exe” to be installed onto the system, which was the hacker’s first point of entry into the victim’s device. From there, the attacker was able to establish a secure and continuous connection between their device, their server, and the infected device through “cryptcat.exe”, “nc.exe”, and “bircd.exe” and they gained root-level access, as is evidenced by their tampering of registry keys. They were likely able to exfiltrate data, such as keystrokes, files, images, documents, etc. to their server, then to their device thanks to their hidden Trojan malware, “poisonivy.exe”. They were able to escalate their privileges and set up tons of other processes-in-hiding to quietly compromise the whole device. They were also able to find the victim’s presumed name, Daniel Faraday, through these compromises. They set up a continuous, encrypted communication channel that kept the connection between the devices ongoing unbeknownst to the victim, even after the original one expired. This was an advanced persistent threat (APV), which I concluded by the nature of the threat being so all-encompassing and data-exfiltration centered. The attacker was trying hard not to get detected and made an effort to keep their connection to the device unbroken. The attacker also spoofed their IP address to another device on the network, “192.168.5.98”, which I determined by connecting the address “poisonivy.exe” communicated with during the capture.

Tools Used

- Volatility
- Websites:
    - Mitre ATT&CK
    - StackExchange
    - Microsoft Forum

Appendix

1a.



1b.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : WinXPSP2x86, WinXPSP3x86 (Instantiated with WinXPSP2x86)
                     AS Layer1 : IA32PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (/mnt/share/KobayashiMaru.vmem)
                      PAE type : No PAE
                           DTB : 0x39000L
                          KDBG : 0x80537d60L
          Number of Processors : 1
     Image Type (Service Pack) : 0
                KPCR for CPU 0 : 0xffdff000L
             KUSER_SHARED_DATA : 0xffdf0000L
       Image date and time : 2018-10-30 20:47:03 UTC+0000
```

2.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem --profile=WinXPSP2x86 pslist
Volatility Foundation Volatility Framework 2.6
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x81fcc800 System                    4      0     54      275 ------      0
0x81f07da8 smss.exe                336      4      3       21 ------      0 2018-10-30 20:46:44 UTC+0000
0x81d2b020 csrss.exe               664    336     12      453      0      0 2018-10-30 20:46:45 UTC+0000
0x81dc4020 winlogon.exe            688    336     25      486      0      0 2018-10-30 20:46:45 UTC+0000
0x819efda8 services.exe            732    688     18      390      0      0 2018-10-30 20:46:45 UTC+0000
0x81b98da8 lsass.exe               744    688     25      339      0      0 2018-10-30 20:46:45 UTC+0000
0x81e92418 vmacthlp.exe            888    732      1       27      0      0 2018-10-30 20:46:45 UTC+0000
0x819edda8 svchost.exe             916    732      9      252      0      0 2018-10-30 20:46:45 UTC+0000
0x81ee5500 svchost.exe             960    732     70      875      0      0 2018-10-30 20:46:45 UTC+0000
0x81d976c8 svchost.exe            1028    732      5       72      0      0 2018-10-30 20:46:45 UTC+0000
0x81e07da8 svchost.exe            1108    732     12      142      0      0 2018-10-30 20:46:46 UTC+0000
0x81e536a0 spoolsv.exe            1308    732     15      189      0      0 2018-10-30 20:46:46 UTC+0000
0x81db4298 hxdef100.exe           1416    732      2       31      0      0 2018-10-30 20:46:46 UTC+0000
0x81d626a0 inetinfo.exe           1432    732     34      540      0      0 2018-10-30 20:46:46 UTC+0000
0x819e2c20 jqs.exe                1464    732      7      214      0      0 2018-10-30 20:46:47 UTC+0000
0x81ede980 cryptcat.exe           1472   1416      1       62      0      0 2018-10-30 20:46:47 UTC+0000
0x81cada80 bircd.exe              1480   1416      2       45      0      0 2018-10-30 20:46:47 UTC+0000
0x81c71508 VMwareService.e        1624    732      2      119      0      0 2018-10-30 20:46:47 UTC+0000
0x81e8f9c0 iroffer.exe            1692   1488      0 --------      0      0 2018-10-30 20:46:47 UTC+0000   2018-10-30 20:46:47 UTC+0000
0x81c85420 iroffer.exe            1728   1692      5       92      0      0 2018-10-30 20:46:47 UTC+0000
0x81df6b20 iroffer.exe            1824   1728      0 --------      0      0 2018-10-30 20:46:47 UTC+0000   2018-10-30 20:46:36 UTC+0000
0x81d32988 wmiapsrv.exe            216    732      5      121      0      0 2018-10-30 20:46:36 UTC+0000
0x819e83c8 wmiprvse.exe            252    916      7      107      0      0 2018-10-30 20:46:37 UTC+0000
0x81edfc18 userinit.exe            368    688      2       34      0      0 2018-10-30 20:46:38 UTC+0000
```

3.

```
0x81a3bc18 explorer.exe            404    368     15      252      0      0 2018-10-30 20:46:38 UTC+0000
0x81d28790 VMwareTray.exe          456    404      1       30      0      0 2018-10-30 20:46:38 UTC+0000
0x81bb3da8 VMwareUser.exe          464    404      5      146      0      0 2018-10-30 20:46:38 UTC+0000
0x81aaa708 jusched.exe             472    404      1       24      0      0 2018-10-30 20:46:38 UTC+0000
0x81e234e8 poisonivy.exe           480    404      1       20      0      0 2018-10-30 20:46:38 UTC+0000
0x81cacda8 msmsgs.exe              488    404      4      127      0      0 2018-10-30 20:46:39 UTC+0000
0x81e579f8 soffice.exe             516    496      1       20      0      0 2018-10-30 20:46:39 UTC+0000
0x81ec6848 soffice.bin             524    516      7      164      0      0 2018-10-30 20:46:39 UTC+0000
0x81c6f7b8 nc.exe                  532    508      1       62      0      0 2018-10-30 20:46:39 UTC+0000
0x81eb3020 winvnc4.exe             548    508      2       81      0      0 2018-10-30 20:46:39 UTC+0000
0x81a2eb78 cmd.exe                 560    508      1       20      0      0 2018-10-30 20:46:39 UTC+0000
0x81b82638 logonui.exe             636    688      4      133      0      0 2018-10-30 20:46:40 UTC+0000
0x81d40418 rundll32.exe            984    404      1       81      0      0 2018-10-30 20:46:43 UTC+0000
```

4.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem --profile=WinXPSP2x86 pstree
Volatility Foundation Volatility Framework 2.6
Name                                                  Pid   PPid   Thds   Hnds Time
-------------------------------------------------- ------ ------ ------ ------ ----
 0x81fcc800:System                                      4      0     54    275 1970-01-01 00:00:00 UTC+0000
. 0x81f07da8:smss.exe                                 336      4      3     21 2018-10-30 20:46:44 UTC+0000
.. 0x81d2b020:csrss.exe                               664    336     12    453 2018-10-30 20:46:45 UTC+0000
.. 0x81dc4020:winlogon.exe                            688    336     25    486 2018-10-30 20:46:45 UTC+0000
... 0x819efda8:services.exe                           732    688     18    390 2018-10-30 20:46:45 UTC+0000
.... 0x81d626a0:inetinfo.exe                         1432    732     34    540 2018-10-30 20:46:46 UTC+0000
.... 0x81db4298:hxdef100.exe                         1416    732      2     31 2018-10-30 20:46:46 UTC+0000
..... 0x81ede980:cryptcat.exe                        1472   1416      1     62 2018-10-30 20:46:47 UTC+0000
..... 0x81cada80:bircd.exe                           1480   1416      2     45 2018-10-30 20:46:47 UTC+0000
```

5.

```
.... 0x81a3bc18:explorer.exe                          404    368     15    252 2018-10-30 20:46:38 UTC+0000
..... 0x81d28790:VMwareTray.exe                       456    404      1     30 2018-10-30 20:46:38 UTC+0000
..... 0x81e234e8:poisonivy.exe                        480    404      1     20 2018-10-30 20:46:38 UTC+0000
..... 0x81bb3da8:VMwareUser.exe                       464    404      5    146 2018-10-30 20:46:38 UTC+0000
..... 0x81aaa708:jusched.exe                          472    404      1     24 2018-10-30 20:46:38 UTC+0000
..... 0x81cacda8:msmsgs.exe                           488    404      4    127 2018-10-30 20:46:39 UTC+0000
..... 0x81d40418:rundll32.exe                         984    404      1     81 2018-10-30 20:46:43 UTC+0000
```

6.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem --profile=WinXPSP2x86 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)  Name                    PID pslist psscan thrdproc pspcid csrss session deskthrd ExitTime
---------- -------------------- ------ ------ ------ -------- ------ ----- ------- -------- --------
0x022e5500 svchost.exe            960 True   False  True     True   True  True    True
0x022de980 cryptcat.exe          1472 True   False  True     True   True  True    True
0x02132988 wmiapsrv.exe           216 True   False  True     True   True  True    True
0x02128790 VMwareTray.exe         456 True   False  True     True   True  True    True
0x01e3bc18 explorer.exe           404 True   False  True     True   True  True    True
0x01f98da8 lsass.exe              744 True   False  True     True   True  True    False
0x021b4298 hxdef100.exe          1416 True   False  True     True   True  True    True
0x02071508 VMwareService.e       1624 True   False  True     True   True  True    True
0x021c4020 winlogon.exe           688 True   False  True     True   True  True    True
0x02207da8 svchost.exe           1108 True   False  True     True   True  True    True
0x01de2c20 jqs.exe               1464 True   False  True     True   True  True    True
0x01dedda8 svchost.exe            916 True   False  True     True   True  True    True
0x01eaa708 jusched.exe            472 True   False  True     True   True  True    True
0x01de83c8 wmiprvse.exe           252 True   False  True     True   True  True    True
0x022c6848 soffice.bin            524 True   False  True     True   True  True    True
0x022dfc18 userinit.exe           368 True   False  True     True   True  True    True
0x0206f7b8 nc.exe                 532 True   False  True     True   True  True    True
0x021626a0 inetinfo.exe          1432 True   False  True     True   True  True    True
0x022b3020 winvnc4.exe            548 True   False  True     True   True  True    True
0x021976c8 svchost.exe           1028 True   False  True     True   True  True    False
0x02140418 rundll32.exe           984 True   False  True     True   True  True    True
0x01f82638 logonui.exe            636 True   False  True     True   True  True    True
0x020ada80 bircd.exe             1480 True   False  True     True   True  True    True
0x020acda8 msmsgs.exe             488 True   False  True     True   True  True    True
0x02085420 iroffer.exe           1728 True   False  True     True   True  True    True
```

7.

```
0x022234e8 poisonivy.exe          480 True   False  True     True   True  True   True
0x01e2eb78 cmd.exe                560 True   False  True     True   True  True   True
0x01defda8 services.exe           732 True   False  True     True   True  True   True
0x02292418 vmacthlp.exe           888 True   False  True     True   True  True   True
0x022579f8 soffice.exe            516 True   False  True     True   True  True   True
0x022536a0 spoolsv.exe           1308 True   False  True     True   True  True   True
0x01fb3da8 VMwareUser.exe         464 True   False  True     True   True  True   True
0x023cc800 System                   4 True   False  True     True   False False  False
0x021f6b20 iroffer.exe           1824 True   False  False    True   False False  False   2018-10-30 20:46:36 UTC+0000
0x0228f9c0 iroffer.exe           1692 True   False  False    True   False False  False   2018-10-30 20:46:47 UTC+0000
0x0212b020 csrss.exe              664 True   False  True     True   False True   True
0x02307da8 smss.exe               336 True   False  True     True   False False  False
```

8.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem --profile=WinXPSP2x86 connscan
Volatility Foundation Volatility Framework 2.6
Offset(P)  Local Address             Remote Address            Pid
---------- ------------------------- ------------------------- ---
0x01e76368 127.0.0.1:1031            127.0.0.1:6667            1728
0x021935e8 127.0.0.1:6667            127.0.0.1:1031            1480
0x021fd550 0.0.0.0:1037              192.168.5.98:3460         480
```

9.

```
root@kali-hunt-01:/mnt/share# volatility -f KobayashiMaru.vmem --profile=WinXPSP2x86 cmdline
Volatility Foundation Volatility Framework 2.6
**************************************************************************
System pid:      4
**************************************************************************
smss.exe pid:    336
Command line : \SystemRoot\System32\smss.exe
**************************************************************************
csrss.exe pid:    664
Command line : C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,3072,512 Windows=On SubSystemType=Windows ServerDll
=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=winsrv:ConServerDllInitialization,2 ProfileControl=Off MaxRequestThreads
=16
```

10.

```
winlogon.exe pid:    688
Command line : winlogon.exe
**************************************************************************
services.exe pid:    732
Command line : C:\WINDOWS\system32\services.exe
**************************************************************************
lsass.exe pid:    744
Command line : C:\WINDOWS\system32\lsass.exe
**************************************************************************
vmacthlp.exe pid:    888
Command line : "C:\Program Files\VMware\VMware Tools\vmacthlp.exe"
**************************************************************************
svchost.exe pid:    916
Command line : C:\WINDOWS\system32\svchost -k rpcss
**************************************************************************
svchost.exe pid:    960
Command line : C:\WINDOWS\System32\svchost.exe -k netsvcs
**************************************************************************
svchost.exe pid:   1028
Command line : C:\WINDOWS\System32\svchost.exe -k NetworkService
**************************************************************************
svchost.exe pid:   1108
Command line : C:\WINDOWS\System32\svchost.exe -k LocalService
**************************************************************************
```

11.

```
hxdef100.exe pid:   1416
Command line : C:\hxdefrootkit\hxdef100.exe hxdef100.ini
**************************************************************************
inetinfo.exe pid:   1432
Command line : C:\WINDOWS\System32\inetsrv\inetinfo.exe
**************************************************************************
jqs.exe pid:   1464
Command line : "C:\Program Files\Java\jre6\bin\jqs.exe" -service -config "C:\Program Files\Java\jre6\lib\deploy\jqs\jqs.conf"
**************************************************************************
cryptcat.exe pid:   1472
Command line : "C:\hxdefrootkit\cryptcat.exe" -L -p 666 -e cmd.exe
**************************************************************************
bircd.exe pid:   1480
Command line : "C:\hidden\bewareircd-win32\bircd.exe"
**************************************************************************
VMwareService.e pid:   1624
Command line : "C:\Program Files\VMware\VMware Tools\VMwareService.exe"
**************************************************************************
iroffer.exe pid:   1692
**************************************************************************
iroffer.exe pid:   1728
Command line : C:\hidden\ir\iroffer.exe
**************************************************************************
iroffer.exe pid:   1824
**************************************************************************
wmiapsrv.exe pid:    216
Command line : C:\WINDOWS\System32\wbem\wmiapsrv.exe
**************************************************************************
```

12.

```
userinit.exe pid:    368
Command line : C:\WINDOWS\system32\userinit.exe
*********************************************************************
explorer.exe pid:    404
Command line : C:\WINDOWS\Explorer.EXE
*********************************************************************
VMwareTray.exe pid:    456
Command line : "C:\Program Files\VMware\VMware Tools\VMwareTray.exe"
*********************************************************************
VMwareUser.exe pid:    464
Command line : "C:\Program Files\VMware\VMware Tools\VMwareUser.exe"
*********************************************************************
jusched.exe pid:    472
Command line : "C:\Program Files\Common Files\Java\Java Update\jusched.exe"
*********************************************************************
poisonivy.exe pid:    480
Command line : "C:\WINDOWS\System32\poisonivy.exe"
*********************************************************************
msmsgs.exe pid:    488
Command line : "C:\Program Files\Messenger\msmsgs.exe" /background
*********************************************************************
soffice.exe pid:    516
Command line : "C:\Program Files\OpenOffice.org 3\program\soffice.exe" -quickstart
*********************************************************************
soffice.bin pid:    524
Command line : "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-quickstart" "-env:OOO_CWD=2C:\\Program Files\\OpenOffice.org 3\\pro
gram"
```

13.

```
nc.exe pid:    532
Command line : C:\inetpub\ftproot\nc.exe  -L -p 6666 -e cmd.exe
*********************************************************************
winvnc4.exe pid:    548
Command line : C:\inetpub\ftproot\VNC4\winvnc4.exe
*********************************************************************
cmd.exe pid:    560
Command line : C:\WINDOWS\system32\cmd.exe  /K C:\Inetpub\ftproot\lock.bat
*********************************************************************
logonui.exe pid:    636
Command line : logonui.exe /status
*********************************************************************
rundll32.exe pid:    984
Command line : C:\WINDOWS\System32\rundll32.exe fldrclnr.dll,Wizard_RunDLL
```

14.

```
Poison Ivy
192.168.5.98
admin
Poison Ivy
)!VoqA.I4-
poisonivy.exe
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders
AppData
V%X_
ALLUSERSPROFILE=C:\Documents and Settings\All Users
APPDATA=C:\Documents and Settings\Daniel Faraday\Application Data
CLIENTNAME=Console
CommonProgramFiles=C:\Program Files\Common Files
COMPUTERNAME=FARADAY
ComSpec=C:\WINDOWS\system32\cmd.exe
HOMEDRIVE=C:
HOMEPATH=\Documents and Settings\Daniel Faraday
LOGONSERVER=\\FARADAY
NUMBER_OF_PROCESSORS=1
OS=Windows_NT
Path=C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH
PROCESSOR_ARCHITECTURE=x86
PROCESSOR_IDENTIFIER=x86 Family 6 Model 12 Stepping 3, GenuineIntel
PROCESSOR_LEVEL=6
PROCESSOR_REVISION=0c03
```

15.

```
abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
ALLUSERSPROFILE=C:\Documents and Settings\All Users
APPDATA=C:\Documents and Settings\Daniel Faraday\Application Data
CLIENTNAME=Console
CommonProgramFiles=C:\Program Files\Common Files
COMPUTERNAME=FARADAY
ComSpec=C:\WINDOWS\system32\cmd.exe
HOMEDRIVE=C:
HOMEPATH=\Documents and Settings\Daniel Faraday
LOGONSERVER=\\FARADAY
NUMBER_OF_PROCESSORS=1
OS=Windows_NT
Path=C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH
PROCESSOR_ARCHITECTURE=x86
PROCESSOR_IDENTIFIER=x86 Family 6 Model 12 Stepping 3, GenuineIntel
PROCESSOR_LEVEL=6
PROCESSOR_REVISION=0c03
ProgramFiles=C:\Program Files
SESSIONNAME=Console
SystemDrive=C:
SystemRoot=C:\WINDOWS
TEMP=C:\DOCUME~1\DANIEL~1\LOCALS~1\Temp
TMP=C:\DOCUME~1\DANIEL~1\LOCALS~1\Temp
USERDOMAIN=FARADAY
```

16.

```
Registry: \Device\HarddiskVolume1\Documents and Settings\Daniel Faraday\NTUSER.DAT
Key name: Run (S)
Last updated: 2010-04-30 01:37:14 UTC+0000

Subkeys:

Values:
REG_SZ        MSMSGS          : (S) "C:\Program Files\Messenger\msmsgs.exe" /background
```

17.

```
SOFTWARE\Classes\http\shell\open\commandV
Software\Microsoft\Active Setup\Installed Components\
Poison Ivy
192.168.5.98
admin
Poison Ivy
)!VoqA.I4-
poisonivy.exe
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders
AppData
V%X_
ALLUSERSPROFILE=C:\Documents and Settings\All Users
APPDATA=C:\Documents and Settings\Daniel Faraday\Application Data
CLIENTNAME=Console
CommonProgramFiles=C:\Program Files\Common Files
COMPUTERNAME=FARADAY
ComSpec=C:\WINDOWS\system32\cmd.exe
HOMEDRIVE=C:
HOMEPATH=\Documents and Settings\Daniel Faraday
LOGONSERVER=\\FARADAY
NUMBER_OF_PROCESSORS=1
OS=Windows_NT
Path=C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH
PROCESSOR_ARCHITECTURE=x86
```

18.

```
root@kali-hunt-01:/mnt/share# strings 480.dmp | grep -Fi "msmsgs.exe"
msmsgs.exe
MSMSGS.EXE
```

19.

```
cmd.exe pid:    560
Command line : C:\WINDOWS\system32\cmd.exe  /K C:\Inetpub\ftproot\lock.bat
```

20.

```
nc.exe pid:     532
Command line : C:\inetpub\ftproot\nc.exe  -L -p 6666 -e cmd.exe
```